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EPRS for Subsea Pipelines Using Combination of FEA and Asset Data Monitoring for Middle East Oil and Gas

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Abstract

This paper examines the integration of Finite Element Analysis (FEA) and data monitoring techniques to develop an Emergency Pipeline Repair System (EPRS) strategy designed explicitly for subsea pipelines within the oil and gas industry. Given the distinct challenges posed by underwater environments, such as high pressure, corrosive conditions, and potential external interference, particularly in shallow water, effective emergency repair strategies are essential for averting environmental catastrophes and maintaining operational continuity.

The primary objective of this paper is to enhance the reliability and efficiency of assessing the needs of emergency repairs for subsea pipelines by implementing advanced analytical methods in conjunction with real-time data monitoring, thereby reducing the risks associated with pipeline failures. This paper will focus on utilising Finite Element Analysis (FEA) to model the structural responses of subsea pipelines under various failure scenarios, including leaks and physical damage. Real-time data monitoring technologies will be utilised to collect critical performance data, enabling proactive assessments and timely interventions during emergencies.

The evaluation of the integrated strategy will employ a thorough methodology, emphasising case studies that simulate realistic emergencies alongside the analysis of actual performance data. Controlled simulations will be conducted to mimic various emergency conditions, allowing for the evaluation of the Emergency Preparedness and Response System (EPRS) under pressure. Key performance indicators will be closely monitored, including response times, repair effectiveness, and the system's ability to minimise environmental impacts. A detailed case study will be documented and analysed to pinpoint strengths and weaknesses within the EPRS framework. Additionally, collecting feedback from participants and stakeholders during the simulations will provide qualitative insights to enhance the quantitative data. This comprehensive approach ensures a meticulous assessment that benchmarks the EPRS against industry safety standards while offering valuable recommendations for ongoing improvements in emergency response strategies.

The proposed EPRS strategy offers several benefits, including enhanced situational awareness, expedited repair processes, and reduced environmental risks. FEA enables engineers to precisely identify failure points

and assess the structural integrity of pipelines before and during emergency repairs. Simultaneously, data monitoring enables the early detection of issues, facilitating immediate action to protect the surrounding marine ecosystem.

In summary, integrating FEA and data monitoring within an Emergency Pipeline Repair System represents a significant advancement in addressing the challenges of subsea pipeline maintenance and enhancing preparedness, thereby fostering safer and more efficient operational practices in demanding underwater environments.

Introduction

Emergency pipeline repair systems (EPRS) play a vital role in ensuring the safety and functionality of pipeline infrastructure, particularly within the oil and gas industry. These systems are specifically engineered to address issues such as leaks, ruptures, or any failures that can endanger public safety, the environment, or the continuity of business operations. The fundamental objective of an EPRS is to establish a high level of preparedness, enabling prompt and efficient emergency repairs when necessary. This involves creating comprehensive guidance and procedural frameworks that streamline the pipeline repair process, thereby ensuring rapid response capabilities.

The EPRS is designed to minimise pipeline downtime, which is crucial for reducing the economic impacts on businesses and communities that rely on continuous pipeline operations. In parallel, safety and environmental considerations are prioritised during the emergency response actions, involving immediate measures to isolate the affected pipeline and secure the surrounding area to prevent further risks.

A critical element for the effective implementation of an EPRS is establishing a detailed procedure and accurately identifying potential damage types. The deployment of an emergency pipeline repair system requires a meticulously coordinated strategy that starts with a comprehensive assessment of the entire pipeline network. This assessment aims to pinpoint vulnerable sections that may be more susceptible to failures. The use of advanced monitoring technologies plays a significant role in this phase, as they can detect anomalies and irregularities in real-time, enabling the quicker identification of potential issues before they escalate.

In the unfortunate event of a pipeline failure, rapid response teams must be swiftly mobilised. These teams should be equipped with a specialised toolkit that includes clamps, patches, welding equipment, and other materials tailored for various pipe sizes and construction materials. Team members must be well-trained in approved repair techniques, understand safety protocols, and be compliant with relevant regulatory standards to ensure safe and effective repairs.

Moreover, contingency plans are crucial in outlining effective communication channels among team members, coordinating with local authorities, and developing strategies to ensure public safety during the repair process. Regular emergency preparedness drills and routine maintenance checks of the pipeline systems and repair equipment are paramount. These practices not only enhance the readiness of response teams but also help boost confidence in their ability to manage crises effectively.

By establishing and executing a comprehensive emergency pipeline repair strategy, oil and gas field operators can significantly reduce risks associated with pipeline failures, minimise operational downtime, and safeguard both the environment and the overall efficiency of their operations. This proactive approach not only protects public health and safety but also ensures the stability of the energy supply chain that is critical to many economies.

The development of the Emergency Pipeline Repair System (EPRS) manual emphasises the integration of digital technologies, particularly the utilisation of asset live data and finite element analysis to assess various damage scenarios. This innovative approach aims to enhance the understanding of potential issues that may arise, guiding effective emergency responses.

Central to this methodology is the analysis of major damage scenarios, leveraging real-time asset data to inform decisions. The finite element assessment enables a detailed examination of the structural integrity of relevant asset components, simulating various conditions and identifying potential vulnerabilities.

In addressing repair technologies, a thorough investigation considers options such as wrapping, repair clamps, and tie-ins. The benefits and drawbacks of each technology are evaluated in the context of the pipelines, taking into account the types of fluids handled, potential risks to personnel, environmental consequences of leaks, and interactions with adjacent facilities, as well as the properties of the pipeline materials.

Furthermore, the EPRS assessment typically extends to Temporary Decommissioning, Isolation, and Recommissioning (TDIR) technologies. Relevant techniques for water treatment, dewatering, flushing, and isolation using double barrier tools are scrutinised. The review examines how these technologies can be applied effectively to the pipeline, with an emphasis on their feasibility and operational impact.

Ultimately, the process culminates in the identification of applicable EPRS solutions tailored to each credible damage scenario. By harnessing advanced digital tools and live asset data, the EPRS aims to deliver informed, timely interventions that effectively address emergencies, ultimately improving safety and operational resilience.

The identification of damage scenarios is crucial for the proper evaluation and selection of TDIR (Temporary Damage Intervention and Repair) and Repair Technologies to be included in the EPRS (Emergency Pipeline Repair System) Solution. The damage scenarios for the pipelines can be categorised into the following clusters:

- **Damage Types:** This is based on the extent of the part that needs to be fixed or replaced.
- **Damage Conditions:** This considers the impacts on pipeline pigability and containment functionality.
- **Damage Locations:** This refers to specific ranges of homogeneous water depths and other field details.

Each scenario will be influenced by a series of local and situational variables, which may affect the repair operations and could limit the applicability of specific options. Variables include seabed conditions, such as slopes, free spans or crossings may impact pipeline stability and the handling performance of specific modules resting on the seafloor.

These considerations influence the ability to correctly position tools and modules, as well as the pipeline's capacity to withstand additional loads imposed by repair tools. Furthermore, it affects the capability of handling tools to secure and move the pipe itself. All of these critical conditions must be accounted for during the development of the EPRS Solution and evaluated thoroughly during the Damage Assessment in the event of an actual damage incident.

Challenges in Current EPRS Strategy Practices

The Emergency Pipeline Repair System (EPRS) is a crucial framework designed to ensure the integrity and safety of pipeline infrastructures. Pipelines transport critical resources such as oil and gas; any failure can result in significant environmental harm and economic loss. Despite its crucial role, EPRS faces several inherent challenges that hinder its effectiveness when responding to emergencies.

One of the primary drawbacks of the EPRS is the prolonged response time associated with emergencies. In many cases, especially in remote or difficult-to-access locations, the time it takes to mobilise repair teams and equipment can exceed acceptable limits. For instance, a leak in an isolated pipeline may not be detected until significant damage has occurred, extending the exposure of hazardous materials to the surrounding environment. Without real-time monitoring technologies, such delays can exacerbate the situation, leading to environmental degradation that could have been avoided. In contrast, implementing advanced monitoring

systems, such as IoT sensors integrated with digital twin technology, can enable the quicker identification of leaks, allowing teams to arrive on-site with the necessary tools in a timely manner.

Another challenge is resource limitations faced by emergency response teams. Effective repairs require not only the right personnel but also specialised equipment tailored to the specific nature of the incident. Additionally, logistical issues, such as accessing heavy machinery in remote areas, can exacerbate these challenges and prolong pipeline downtime, leading to financial losses and reduced service reliability.

In many emergencies, responders also contend with incomplete data about the pipeline's operational status. Without comprehensive information—such as pressure, flow rates, and historical maintenance records—teams might make decisions based on limited insights, potentially jeopardising the overall safety and effectiveness of the repair efforts. For example, if a team is unaware of previous stress points in a gas pipeline, their repair approach may overlook critical factors, leading to further incidents. The lack of data-driven decision-making highlights the need for improved monitoring solutions that provide accurate, real-time insights into pipeline conditions.

Moreover, human error remains a significant concern during emergency responses. High-stress environments and the urgency of the situation can lead to misjudgments or oversights by operators. Factors such as fatigue and inadequate training can exacerbate the risk of errors during critical repairs. For instance, a worker may misinterpret pressure readings or overlook essential safety protocols, which can result in potential safety hazards or additional damage to the pipeline infrastructure. Addressing this issue requires comprehensive systems that support decision-making to minimise the likelihood of human error.

The high costs associated with emergency repairs can be staggering. Unplanned incidents typically involve not only immediate repair costs but also long-term financial repercussions, including environmental cleanup efforts and regulatory fines. When an oil pipeline suffers a spill, for example, the costs related to containment, cleanup, and legal ramifications can quickly escalate into millions of dollars. Additionally, the effect on service disruption can negatively impact operational revenues. Thus, investing in preventative measures and advanced technologies can mitigate these financial burdens while enhancing pipeline integrity.

Furthermore, there are significant environmental risks associated with unplanned pipeline incidents. Rapidly conducted emergency responses often prioritise immediate containment, neglecting comprehensive long-term evaluations of the ecological impacts. A pipeline leak occurring in sensitive areas, such as wetlands, can have devastating effects on local wildlife and flora. Addressing these potential consequences demands a more balanced approach to emergency responses—one that not only focuses on immediate repairs but also considers the ongoing health and sustainability of the surrounding environment.

EPRS Strategy Enhancement using Finite Element Analysis of Damages

Finite Element Analysis (FEA) is a pivotal tool that significantly enhances the Emergency Pipeline Repair System (EPRS), particularly during critical events such as anchor drag. This sophisticated analytical method enables engineers to simulate and thoroughly assess the structural integrity of pipelines under a wide range of loading conditions and potential impacts. By creating detailed models of the pipeline and analysing its interactions with external forces—such as the dragging motion of anchors—FEA enables the identification of stress concentrations and potential failure points that may remain hidden through conventional analytical techniques. This proactive and vigilant approach ensures that pipelines are equipped to endure the challenging dynamic forces associated with anchor drag events, thereby helping to prevent catastrophic failures and ensuring safer operational practices.

In addition, FEA allows for a comprehensive examination of the pipeline's response to various anchor drag scenarios, taking into account an array of factors, including soil interaction, material properties, and environmental conditions. By varying these parameters within the simulations, engineers can uncover critical insights into possible outcomes and devise tailored strategies to mitigate risks effectively. This capability is invaluable not only during the planning and designing phases of pipelines but also plays a

crucial role in shaping emergency response strategies. It helps teams identify the most vulnerable points in the system, enabling them to prioritise repair efforts during actual emergencies based on informed assessments.

Moreover, the knowledge gained from FEA can significantly elevate the level of training and preparedness among emergency response personnel. By integrating the outcomes of FEA simulations into training programs, teams can engage in immersive, scenario-based exercises that deepen their understanding of the potential consequences of anchor drag events on pipelines. This practical experience equips responders with essential skills and knowledge, enabling them to execute repair strategies with confidence and efficiency even in high-pressure situations. Ultimately, incorporating Finite Element Analysis into the EPRS framework not only fosters a culture of safety and resilience but also promotes an environment of continuous improvement in the management and maintenance of pipeline systems.

EPRS Strategy Enhancement using Digital Twin (Asset Data Monitoring)

Integrating digital twin technology into the EPRS paradigm offers a promising solution to overcome these challenges. A digital twin is a virtual representation of a physical asset, incorporating real-time data, historical records, and predictive analytics. This technology can revolutionise how pipeline emergencies are managed, improving efficiency and safety.

First and foremost, instantaneous asset data monitoring through digital twins can drastically reduce response times. By continuously tracking the pipeline's conditions with IoT sensors, anomalies can be detected in real time, allowing operators to act swiftly when a potential issue arises. For example, if a digital twin forecasts a drop in pressure indicating a possible leak, the system can immediately alert the relevant teams for prompt action, thereby minimising the risk of extensive damage and environmental impact.

Pipeline data monitoring plays a pivotal role in ensuring the safety and integrity of offshore oil and gas transportation systems. Given the harsh environmental conditions and technical complexities associated with underwater pipelines, having a robust emergency repair system bolstered by reliable data monitoring is crucial. This involves the use of advanced technologies to continuously collect and analyze data related to the operational health of the pipeline.

One of the key components of pipeline data monitoring is the integration of various sensor technologies. For example, pressure sensors can be installed along the pipeline to measure internal pressure fluctuations. Monitoring pressure is vital as it can indicate blockage, leaks, or ruptures [3,4].

In addition to pressure sensors, temperature sensors play a crucial role in detecting temperature variations that can signal a problem. For instance, a sudden drop in temperature might suggest a leakage where the fluid is escaping into the environment. Data from these sensors can be transmitted in real time to a central monitoring station where algorithms analyze the data for outliers and deviations from normal operating conditions.

Flow rate monitors also contribute significantly to pipeline data monitoring. By analyzing the flow rate of oil or gas through the pipeline, operators can identify irregular patterns potentially indicating blockages or leaks. For instance, if the flow rate significantly decreases without a corresponding change in demand, it could signal a breach in the pipeline. Monitoring systems can alert operators, enabling them to act swiftly and assess the pipeline for potential issues [2].

The implementation of machine learning algorithms enhances the capability of data monitoring systems. These algorithms can analyze historical data to predict potential failures based on patterns observed over time. If the data indicates that specific sections are deteriorating faster than others, proactive maintenance can be scheduled before a significant issue arises, thereby reducing the likelihood of emergency repairs.

Moreover, the integration of automated shut-off valves can act as a fail-safe mechanism. In the event of a detected leak or pressure anomaly, these valves automatically shut down the flow, preventing further

escalation of the situation and mitigating environmental risks. This proactive approach allows for safer operations and minimizes the impact of any incidents.

The predictive capabilities offered by digital twins allow organisations to engage in advanced predictive analytics. By analysing historical data alongside real-time metrics, these systems can anticipate potential failures before they occur. For instance, if data reveals that specific segments of a pipeline experience repeated stress, maintenance can be scheduled proactively, thus reducing the number of emergencies and enhancing overall pipeline reliability.

Digital twins also facilitate comprehensive data integration, presenting a holistic view of the pipeline's condition. This capability empowers response teams with detailed insights that enhance their decision-making during emergencies. If a significant anomaly is detected, the digital twin can provide data on potential interconnected issues, supporting a more informed and effective repair strategy.

One of the conceptual digital twin models for subsea pipelines was proposed by Bhowmik [1, 5]. The framework integrates a machine learning (ML)-based predictive maintenance system with computational modeling. Initially, a finite element (FE) model is calibrated using real sensor data from the pipeline to generate baseline operational behavior. A trained ML model then predicts potential anomalies, including high pressure fluctuations, slugging, or leaks, based on variations in pipeline conditions in Figure 1. This approach offers promise for reducing maintenance costs through proactive decision-making and can be used for used in enhanced EPRS strategy development.

EPRS Elements Structure

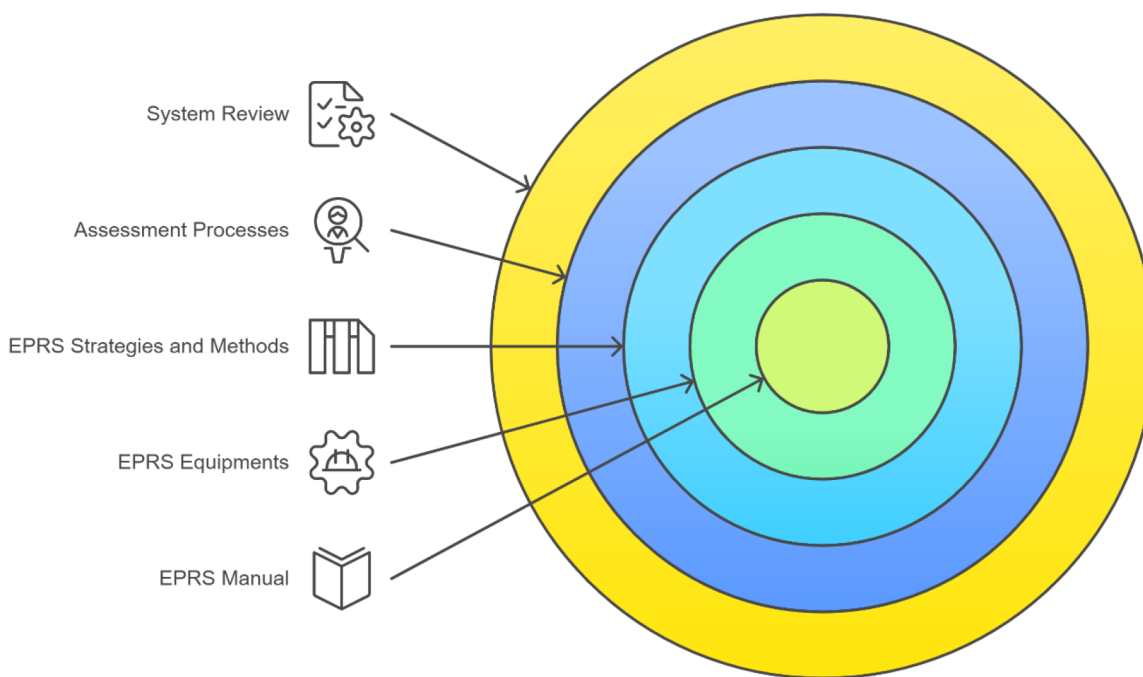


Figure 1—Traditional EPRS Strategy

EPRS stages range from reactive to proactive strategies.



Figure 2—EPRS Stages from reactive to proactive

The modified Digital Twin (DT) architecture from Figure 3 for EPRS has been shown in Figure 3.

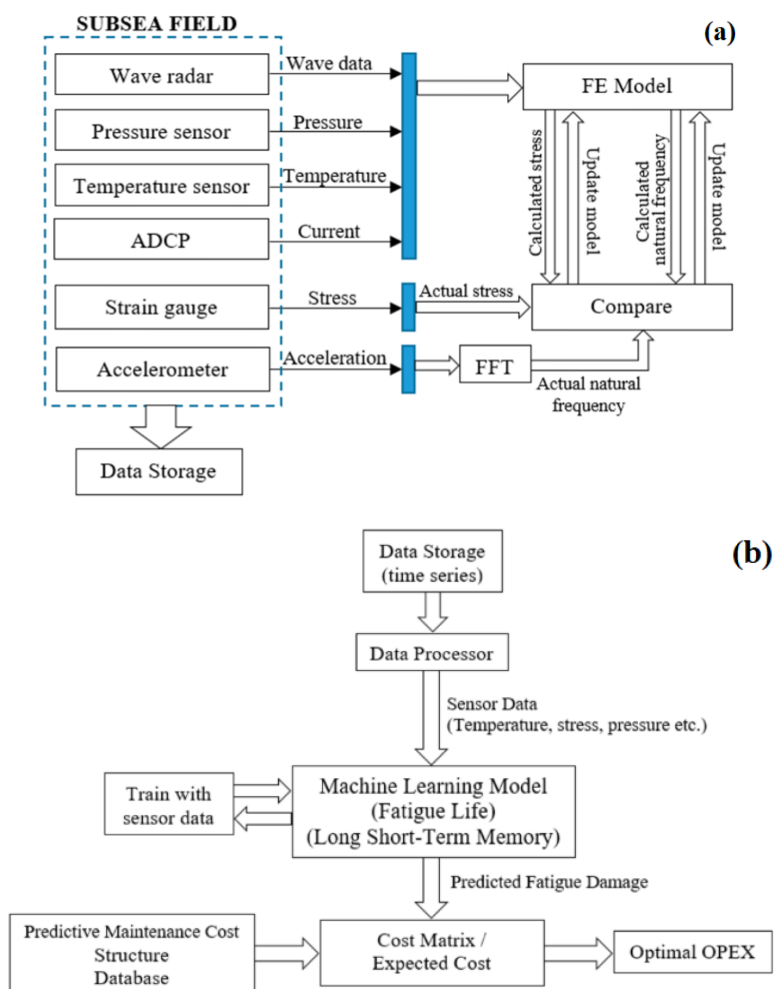


Figure 3—Conceptual design of a DT for subsea pipeline proposed by Bhowmik [1]. (a) Calibrated FE model compared to field sensor data, and (b) data-driven predictive maintenance model

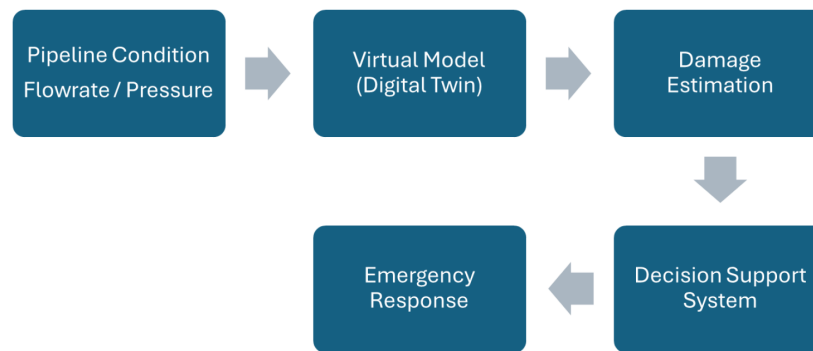


Figure 4—Digital Twin Architecture using Asset Data Monitoring to support

Case Study: Example of an offshore Pipeline dragged by Anchor on Seabed and Crossing

A 32-inch pipeline with lateral drag from an anchor is used as a case study to evaluate potential damage. The finite element (FE) model helps determine if repairs are necessary. For instance, a lateral drag of 55 meters is acceptable in areas without buckling, while a lateral drag of 75 meters is permissible in areas where buckling occurs.

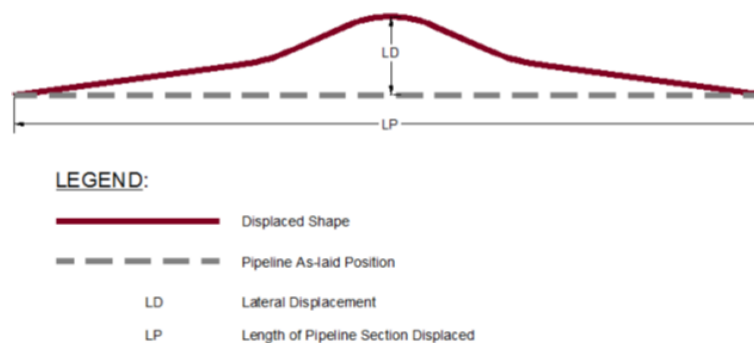


Figure 5—Case: Lateral drag of 32" Pipeline by Anchor

Models have been developed for each pipeline to consider what the pipeline would look like after a pullover event. The models assume that the pipelines remain pressurised following the pull over.

The steps in the analysis are summarized below.

Step 1: Define the pipe geometry, material properties, contact surfaces and apply gravity

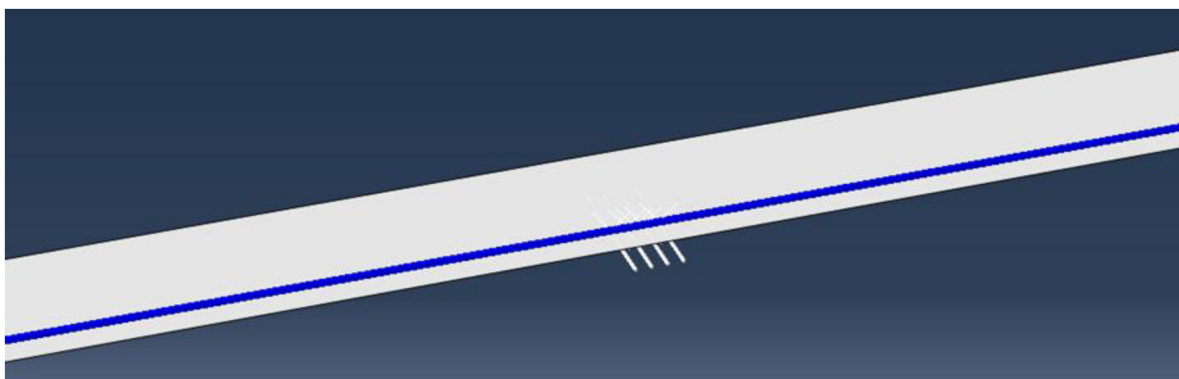


Figure 6—Pipeline Resting on the Seabed

Step 2: Initialise model –apply the normal operating pressure and temperature.

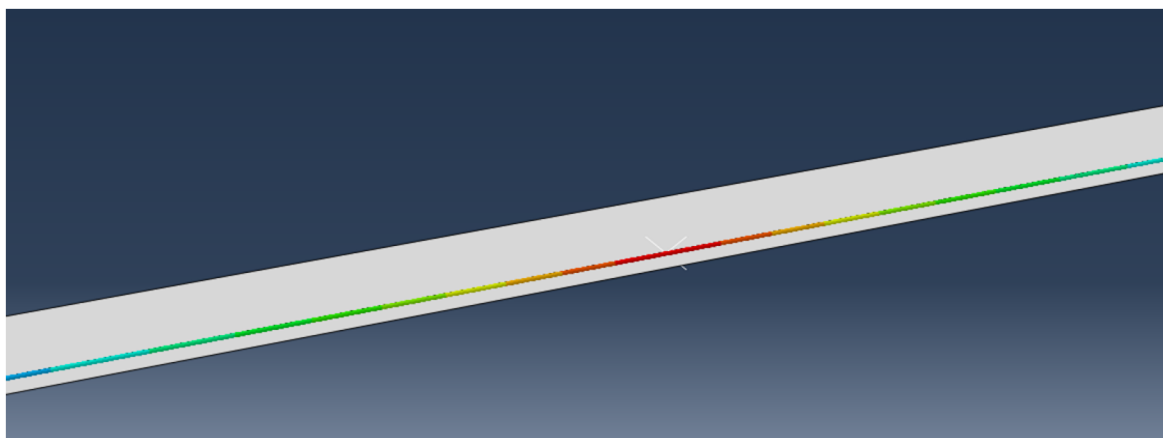


Figure 7—Temperature and pressure added to the model

Step 3: Pull the pipeline laterally similar to anchor drag

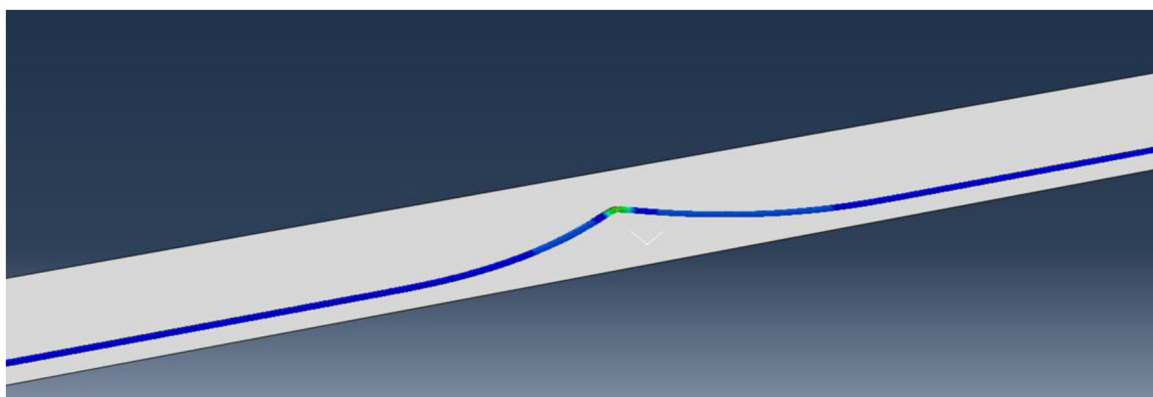


Figure 8—Pipeline laterally displaced through anchor drag

The Abaqus finite element (FE) model was developed to simulate pullover damage. A suitable length of section of pipeline were modelled so that it will be sufficient for the built in force to develop within the model, and for axial movement to reduce to zero before the model ends. Pipe beam elements were used to model the longitudinal pipeline behavior. Cross sectional effects are small and not included.

FEA modelling has increasingly been used to assess pipeline damages. Some of the damages have been validated against full scale test results. Moreover, guidance on the analysis procedure is also available. However, for complex damages such as impact due to drag, the FEA methods have not been proven via full scale testing. To ensure quick decision-making, it is recommended to model displacement scenarios in advance. This approach helps identify situations where the impact of displacement needs to be considered and where it does not.

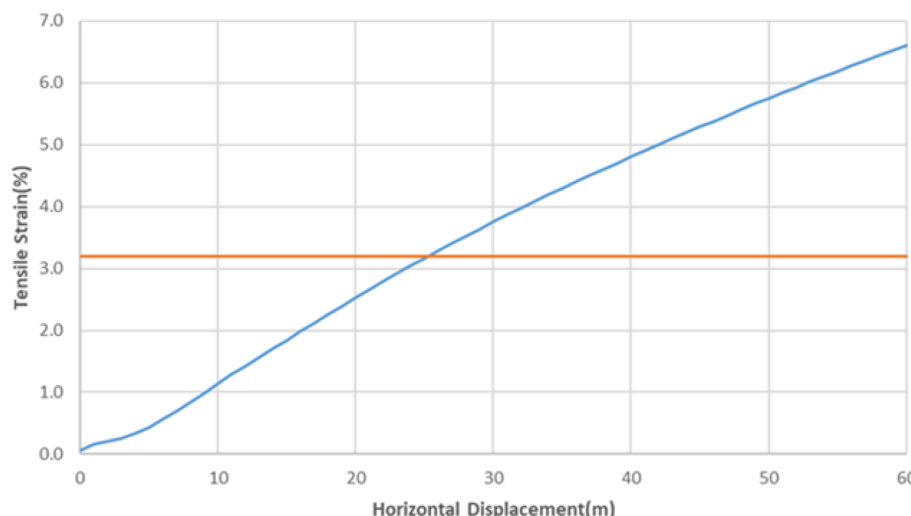


Figure 9—Strain developed due to drag and Strain limit

Conclusion

The finite element method (FEM) and asset data monitoring are vital tools that can significantly enhance emergency pipeline repair strategies. FEM offers advanced modelling capabilities that allow engineers to simulate various stress conditions on pipeline structures. A detailed finite element analysis helps predict potential failure points, assess the extent of any damage, and provide valuable insights into the pipeline's structural integrity. By identifying critical areas requiring immediate attention, engineers can prioritise repairs and allocate resources more effectively.

On the other hand, asset data monitoring involves the continuous collection and analysis of key performance metrics, such as flow rates, pressure levels, temperature variations, and environmental factors. By monitoring these parameters in real-time, operators can detect anomalies or changes that may indicate potential failures or breaches before they escalate into serious issues. When these insights are combined with FEM analysis, pipeline operators can adopt a proactive maintenance approach, enabling timely interventions that prevent significant damage and downtime.

Together, these technologies enable pipeline operators to make rapid, informed decisions in emergencies. Utilising predictive analytics based on historical data enables quick assessments of pipeline conditions and facilitates the efficient deployment of repair teams where they are most needed. By integrating FEM and asset data monitoring, operators enhance the resilience of pipeline systems, improve safety measures, and minimise the financial impacts associated with emergency repairs. This comprehensive approach ultimately reinforces the reliability of the pipeline network in the face of unforeseen challenges.

In summary, the Emergency Pipeline Repair System, while critical, faces several challenges that can hinder its effectiveness during emergencies. However, by harnessing the capabilities of finite element analysis and Asset data monitoring through digital twin technology, operators can significantly enhance their EPRS strategies with minimal EPRS components. By ensuring real-time monitoring, predictive maintenance, efficient resource management, and comprehensive training, digital twins represent a transformative opportunity for safer and more effective pipeline management. This integration not only enhances operational integrity but also protects the environment, resulting in a more sustainable and resilient infrastructure.

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